Undergraduate Student's Topology(H) Course
An Introduction

Zuoqin Wang

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Outline

1. What is topology...
   - A: The origin of the subject
   - B: Some abstract words
   - C: Some concrete pictures
2. Some topological theorems that we have learned/will learn
   - Part I: Point-Set Topology and Applications to Analysis
   - Part II: Algebraic/Geometric Topology

What's next ☞

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A: The origin of the subject/name

Euler!

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He used the name

Analysis situs

Leibniz

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(but what Leibniz had in his mind could be an entirely different thing ...)

Listing

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He combined two Greek words

- topos($\tau o \pi o \varsigma$): place, location
- logos($\lambda o \gamma o \varsigma$): study

Mobius

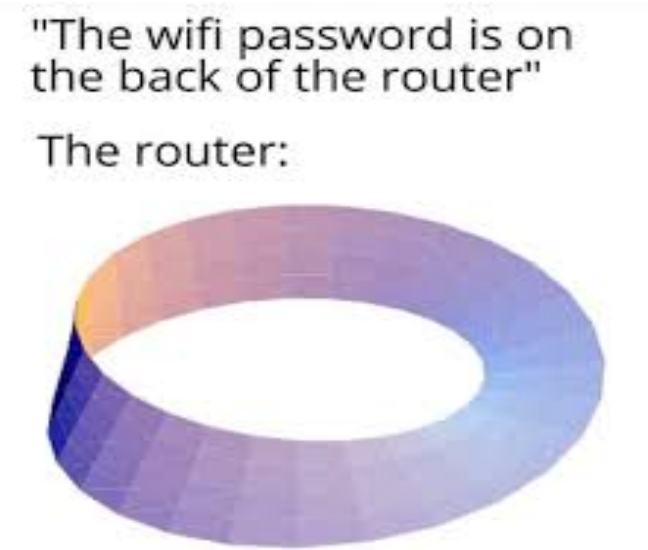
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Mobius

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But Listing went further in exploring the properties of strips with higher-order twists instead of the non-orientability ...

Gauss!

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Gauss also made great contributions to topology:

- the linking number/integral $\rightsquigarrow$ knot theory, DNA ...
- Gauss-Bonnet-Chern theorem: curvature and topology

Riemann!

- Another student of Gauss: Bernhard Riemann (1826-1966)



Riemann!

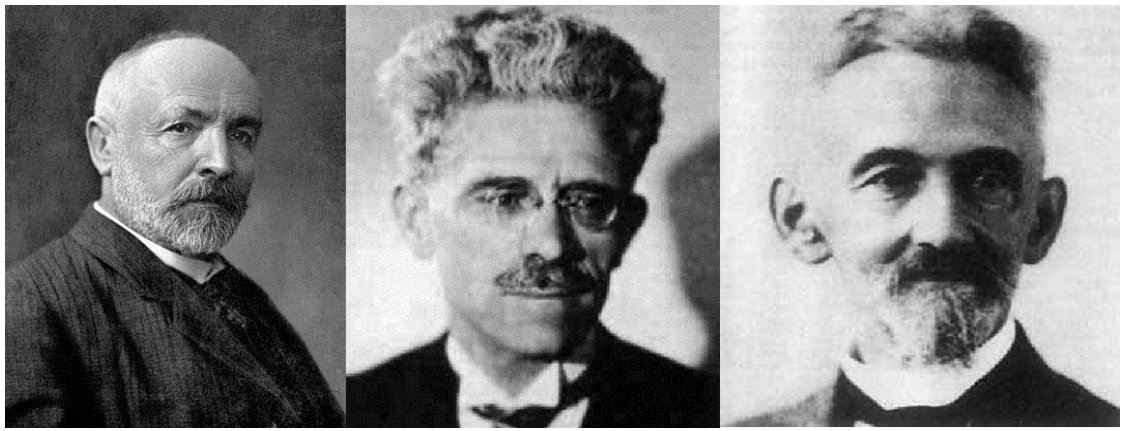
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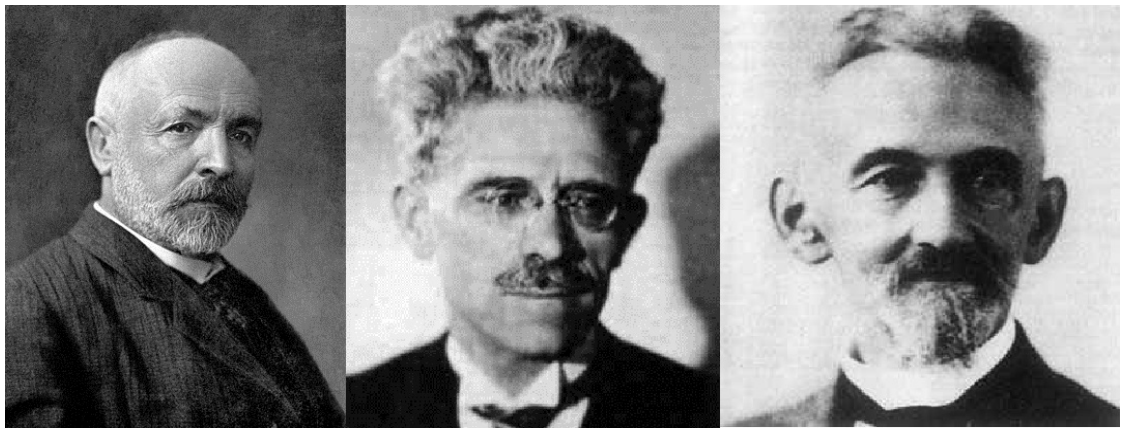
who also made great contributions to topology:

- Riemann surface $\rightsquigarrow$ Topological methods in complex function theory
- Manifolds $\rightsquigarrow$ the conception of abstract/higher-dim spaces
- Incorrect use of the Dirichlet Principle $\rightsquigarrow$ compactness

Cantor, Frechet, and Hausdorff



Cantor, Frechet, and Hausdorff



In a totally different direction,

- Georg Cantor(1845-1918): abstract set theory $\rightsquigarrow$ limit points, ideas related to convergence of sequences(1870s)
- Maurice Frechet(1878-1973): axioms for convergence, and axioms for metric spaces (1905)
- Felix Hausdorff(1868-1942): axioms for topological spaces (Neighborhood approach)(1912)

Question: What is topology?

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Of course I can't give a simple answer to this question now. You should have your own answer at the end of this semester. Here let's see how mathematicians say: